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PROJECT AND TEAM INFORMATION

Project Title

SkillArc

Team Name: *Team #*	*Apex*
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PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

The SkillArc application is crafted to make learning easier for students. SkillArc students are expected to select a topic. Move on from there and explore different topics in order. Students are mandated to begin with the basic subjects and advance to the more difficult ones. Notes, videos, quizzes, and code challenges will be provided for each subject to help students get to know the topics. Students are able to assess their coding proficiency through platforms like Leetcode. SkillArc will include scores, contest leaderboards, and rewards, which will make learning more fun thereby encouraging Students should practice more. Pupils are able to exchange their notes. Use resources that could prove beneficial for others applying. The main purpose of developing the SkillArc app is to simplify learning concepts in data science, object-oriented programming, and other topics, making them interactive and enjoyable students. Now that the material is centralized in one location, students don't have to go through numerous sites for their studies. Learning can be personalized according to their speed, and they can assess their own progress.

Updated Project Approach and Architecture (2 pts)

Part	Technology / Tech Stack	Description
DSA Implementation	C	Arrays, Linked Lists, Stacks, Queues, Graphs, Hashing, Searching and Sorting will be used for managing learning content, questions, student records, and leaderboard data.
C++ OOP Implementation	C++	Classes will be created for major entities such as Student, Subject, Topic, Quiz, Contest, Note, and Reward using OOP concepts and file handling.
Platform Interface	HTML, CSS, JavaScript, Python, Streamlit	The frontend and interactive interface will provide subject-wise roadmaps, study material, quizzes, coding practice, progress tracking, note sharing, contests, leaderboards, Credit Points, and rewards.
Data & Integration	SQLite, JSON, Git/GitHub	SQLite or JSON will be used for storing project data, while Git/GitHub will be used for managing and integrating the project code.



Tasks Completed (7 pts)

Task Completed	Team Member
• Created separate folders for Arrays, Stack, and Queue. • Added basic concept notes, README files, and YouTube learning resources. • Organized DSA topics into topic-wise folders. • Prepared the structure for future practice questions and quizzes.	Samiksha Gurung
• Set up the development environment using Python, Streamlit, and GitHub. • Developed a dashboard for student information, credit points, progress, and streaks. • Implemented sidebar navigation between application sections. • Maintained Git commits to track development progress and individual contributions.	Vaibhavi Thapliyal
• Created the initial project repository structure and development folders. • Configured the C++ compiler and tested it in the development environment. • Created a quiz prototype using Student and Question classes. • Implemented question display, answer validation, and score and credit-point updates.	Tanisha Rai

Challenges/Roadblocks (7 pts)

- **Integration of Modules:** Connecting the DSA content, C++ backend, and user interface in a consistent way is a challenge. We plan to use a common structure for topics and questions so that all modules work together properly.
- **Managing Quiz Questions and Scores:** Making sure the scores and credit points are added correctly when there are questions is something we need to fix. We will test how the quiz logic works with questions and answers to see if everything adds up properly
- **Saving Student Progress:** : It's important that students don't lose their progress when they close the program. We are planning to use a database to keep track of their scores the topics they have studied and the credit points they have earned.
- **Quiz Navigation:**Students should be able to skip questions move forward and go back, to ones if needed. We will add these navigation options. Test them to make sure the quiz works smoothly.
- **User-Friendly Interface:** The interface must clearly show the topics, questions, scores and progress in a way that is easy to understand. We will improve how things look and feel and test the interface with parts of the system to make sure the platform is easy to use.

Tasks Pending (7 pts)

Task Pending	Team Member
• Implement student registration and store their details, as well as, quiz results • Enhance the quiz system to include multiple questions, answer validation and final score calculation. • Classify questions into topics and integrate DSA related information into quiz. • Implement navigation features in order to skip questions and go back to them later. • Test quiz functionality and fix identified issues.	Samiksha Gurung
• Improve the C++/OOP components and complete the quiz flow with proper score calculation and result feedback. • Connect the backend with the other project modules. • Support persistent storage of student details and results. • Perform functional and integration testing and fix errors.	Tanisha Rai
• Connect the interface with the backend and student data. • Display learning progress, completed topics, and credit points. • Develop the leaderboard, reward cards, and achievement badges. • Add topic-wise notes and learning resources. • Improve the interface layout, accessibility, and interactive elements. • Test the interface and fix navigation and integration issues.	Vaibhavi Thapliyal

Project Outcome/Deliverables (2 pts)

- **Subject-wise Roadmaps:** We believe that the subject-wise roadmaps provide learning paths to students in different fields like DSA,programming.Subject-wise roadmaps offer the option to categorize every topic in a specific order, so students can follow without getting confused.
- **Student Dashboard:** The Student Dashboard is our feature that provides an overview of completed topics coding activities,Credits, achievements, and overall progress.The student dashboard makes it convenient for students and teachers to monitor students' progress
- **Credit-Based Reward System:** The Credit-Based Reward System offers students the opportunity to earn credits through codingHaving trouble keeping up with streaks, creating, and sharing notes, and helping peers.Credit-based reward system is a way to acknowledge students' effort and reward them with credits for further incentive.
- **Streak and Achievement Tracking:** A tool for tracking learning and coding streaks, along withAchievements to keep students motivated.Streak and achievement tracking helps students stay updated on their progress and practice regularly.
- **Notes and Peer-Learning Module:** This platform allows students to add, share, and access notes and peer-learning resourcesnotes and learn together effectively.The Notes and Peer Learning Module is a way to engage students in effective teamwork.Supports students in learning from one another.
- **Progress Reports:** This format is used to document student activities, track credits, monitor streaks and achievements for evaluating performance.The Progress Report serves to evaluate student advancement and offer help when required

Progress Overview (2 pts)

- **Completed So Far:** The group has established the primary project repository and categorized DSA topics such asCreate separate folders for Arrays, Stacks, and Queues, and include basic notes and learning resources in each.The C++ compiler has been set up and validated, and an initial prototype for a quiz has been crafted with Student and Questionclasses.
- **Ahead of Schedule:** The compiler setup and initial quiz prototype have provided a good starting point for back-end development.
- **Pending Work:** The user interface is yet to be developed.Connecting the DSA content with the quiz module, saving student progress, and adding better quiz navigation are also pending.
- **Next Phase:** The group will concentrate on crafting the interface, linking all three components, and verifying the platform before proceedingEverything is running smoothly

Codebase Information (2 pts)

Repository: `https://github.com/samiksha1897-bot/Skill-Arc-PBL`

Branch: main

Important Commits and Updates:

- Organized DSA topics into Arrays, Stack, and Queue folders with notes, README files, and learning resources.
- Started structuring OOP modules to organize concepts and support future integration.
- Set up the initial interface and development environment using Python and Streamlit.
- Prepared the project structure for integrating quizzes and practice questions.
- Organized project files to support collaboration and future development.

Testing and Validation Status (2 pts)

Test Case	Type	Status	Notes
Valid username and name	Positive	Pass	Accepts valid characters
Invalid username and name	Negative	Fail	Rejects invalid characters
Valid email format	Positive	Pass	Accepts .com or .in addresses
Invalid email format	Negative	Fail	Rejects incorrect email formats
Strong password	Positive	Pass	Meets password requirements
Weak password	Negative	Fail	Should enforce password rules
Valid login credentials	Positive	Pass	Login successful
Invalid login credentials	Negative	Fail	Should display an error message
Empty required fields	Negative	Fail	Should show a validation message

Deliverables Progress (2 pts)

Key Deliverable	Details	Status
Tasks Completed	Organized project and DSA resources for Arrays, Stacks,and Queues into folders. Developed a basic C++ quiz prototype with answer checking and credit-point updates.	Completed
Overall Progress	the initial DSA content and basic C++ quiz functionality	In Progress
Student Dashboard	show student information, credit points, progress, and current streaks.	In Progress
Topic-wise Learning	resources for Arrays, Stacks and Queues.We need notes, questions and quizzes.	In Progress
Quiz and Navigation	basic answer validation and scoring are already implemented.Question skipping, revisiting and extra questions .	In Progress
Data Storage	student registration and persistent saving of student details scores and progress .	In progress
Interface Integration	connecting the Streamlit interface to the DSA content and the C++ backend.	Pending
Rewards and Progress Tracking	complete credit-point displays, achievements and current progress tracking reports.	In progress
Testing and Validation	Test quizzes, navigation, data storage, and interface integration.	Pending
Expected Final Deliverable	SkillArc platform, with roadmaps, quizzes, coding practice progress tracking and a user-friendly interface.	Pending